

3-e Factorial in Prolog

PROGRAM:

```
factorial(0, 1).
factorial(N, R) :-
    N > 0,
    N1 is N - 1,
    factorial(N1, R1),
    R is N * R1.

run_factorial :-
    write('Enter a number: '), read(N),
    factorial(N, R),
    format('Factorial of ~w = ~w~n', [N, R]).
```

OUTPUT:

```
SWI-Prolog console
File Settings Tools Debug Help
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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

22 ?- consult(['c:/Users/kames/OneDrive/Documents/factorial.pl']).
true.

23 ?- run_factorial.
Enter a number: 0.
Factorial of 0 = 1
true .

24 ?- run_factorial.
Enter a number: 4.
Factorial of 4 = 24
true .

24 ?- run_factorial.
Enter a number: -5.
false.

24 ?- ▲
```